

COSTA MESA WAYNE, CA, USA

WMO: 722977

Lat: 33.680N

Lon: 117.866W

Elev: 17

StdP: 101.13

Time Zone: -8.00 (NAP)

Period: 99-19

WBAN: 93184

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
12	6.0	7.2	-14.0	1.1	19.5	-9.6	1.7	19.7	9.5	17.0	8.3	17.7	0.6	30	0.345

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
8	7.3	32.1	18.5	30.2	18.5	28.4	18.3	21.8	27.9	21.2	26.8	20.6	25.9	3.5	230

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
19.8	14.6	24.6	19.0	13.9	24.0	18.5	13.4	23.5	63.7	27.9	61.3	26.8	59.3	26.0	25.4

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max		
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
7.2	5.8	5.4	DB	3.4	37.2	1.4	2.8	2.4	39.3	1.6	40.9	0.8	42.5	-0.2	44.5	(4)
			WB	0.6	23.1	1.2	0.8	-0.3	23.7	-1.0	24.2	-1.7	24.6	-2.5	25.2	(5)

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6)	Temperatures, Degree-Days and Degree-Hours	DBAvg	18.2	14.4	14.4	15.6	16.7	18.3	20.0	22.3	22.8	22.4	20.3	17.1	14.1	(6)
		DBStd	3.95	2.84	2.69	2.39	2.42	2.10	1.67	1.88	1.87	2.56	2.67	2.74	2.41	(7)
		HDD10.0	3	1	1	0	0	0	0	0	0	0	0	0	1	(8)
		HDD18.3	614	126	114	92	61	23	2	0	0	9	54	133	(9)	(9)
		CDD10.0	3004	138	123	172	201	257	299	382	397	373	318	214	128	(10)
		CDD18.3	574	5	3	6	12	22	52	124	139	123	68	18	2	(11)
		CDH23.3	2992	74	52	79	132	138	127	467	619	673	434	171	26	(12)
(14)	Wind	CDH26.7	847	14	12	21	44	56	25	91	142	227	159	52	3	(13)
		WSAvg	2.2	1.7	2.0	2.2	2.5	2.7	2.7	2.6	2.4	2.1	1.8	1.6	1.6	(14)
		PrecAvg	269	62	68	55	21	3	1	1	3	9	5	31	37	(15)
		PrecMax	663	200	316	253	103	12	12	5	44	49	26	134	106	(16)
		PrecMin	90	0	0	0	0	0	0	0	0	0	0	0	1	(17)
		PrecStd	149	58	76	58	28	3	2	1	9	14	6	31	32	(18)
		DB	28.7	28.7	29.2	31.7	32.5	29.9	31.2	32.2	35.1	34.5	31.4	26.2	(19)	(19)
(20)	Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	MCWB	13.3	13.8	14.5	15.7	15.4	20.0	20.8	20.1	20.5	17.2	15.6	12.1	(20)	(20)
		DB	25.3	24.5	25.4	27.1	27.0	26.2	28.9	30.1	31.6	30.8	27.8	23.0	(21)	(21)
		MCWB	12.1	12.7	13.3	14.4	16.2	19.1	20.2	20.4	20.5	16.5	14.2	12.7	(22)	(22)
		DB	22.9	21.9	22.4	23.6	23.5	24.4	27.4	28.1	29.1	27.8	24.9	21.1	(23)	(23)
		MCWB	12.0	12.6	13.2	14.1	16.8	18.4	20.3	20.5	20.1	16.8	13.8	12.0	(24)	(24)
		DB	20.3	19.3	20.0	21.0	21.6	23.1	25.9	26.6	27.1	25.3	22.3	19.0	(25)	(25)
		MCWB	11.8	12.4	13.5	14.3	16.1	17.8	19.9	20.1	19.8	17.0	13.9	12.1	(26)	(26)
		WB	16.3	16.5	17.1	17.5	19.0	20.6	22.6	22.5	22.8	21.5	17.9	16.8	(27)	(27)
		MCDB	21.1	21.2	23.8	24.9	25.1	26.8	28.5	28.8	30.4	27.0	23.8	20.7	(28)	(28)
		WB	15.2	15.3	15.9	16.3	18.0	19.6	21.5	21.6	21.7	20.0	17.0	15.5	(29)	(29)
(30)	Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	MCDB	18.8	19.5	20.7	22.2	23.4	25.5	26.4	27.5	28.8	25.7	22.0	18.6	(30)	(30)
		WB	14.5	14.6	15.1	15.5	17.1	18.9	20.8	20.9	21.0	19.1	16.3	14.7	(31)	(31)
		MCDB	18.2	18.4	19.5	20.8	21.9	23.9	25.8	26.4	26.9	24.4	21.0	17.8	(32)	(32)
		WB	13.7	13.8	14.4	14.9	16.5	18.2	20.2	20.4	20.4	18.2	15.5	13.9	(33)	(33)
		MCDB	17.6	17.6	18.3	19.6	20.8	22.4	24.9	25.5	25.7	23.0	20.0	17.1	(34)	(34)
		MDBR	10.1	9.1	8.3	7.9	6.8	6.4	6.8	7.3	8.1	8.7	10.0	9.8	(35)	(35)
		5% DB	15.0	14.1	13.6	13.5	10.9	8.5	8.8	9.6	11.5	14.4	14.9	13.6	(36)	(36)
		MCWBR	6.6	6.3	5.9	5.4	4.4	3.5	3.1	3.4	4.2	5.7	6.4	6.9	(37)	(37)
		5% WB	10.2	9.7	9.0	9.4	8.6	8.3	7.9	8.5	9.9	10.3	10.3	8.9	(38)	(38)
		MCWBR	5.5	5.2	4.7	4.4	3.7	3.5	3.0	3.3	3.9	4.7	5.3	5.4	(39)	(39)
(40)	Clear-Sky Solar Irradiance	taub	0.323	0.333	0.357	0.363	0.362	0.377	0.383	0.368	0.353	0.354	0.312	0.311	(40)	(40)
		taud	2.537	2.522	2.430	2.420	2.454	2.416	2.466	2.493	2.526	2.499	2.609	2.611	(41)	(41)
		Ebn at Noon	893	920	919	924	923	904	896	908	911	883	902	891	(42)	(42)
		Edh at Noon	82	93	109	116	113	117	111	106	98	93	76	73	(43)	(43)
(44)	All-Sky Solar Radiation	RadAvg	3.20	4.08	5.45	6.62	7.08	7.43	7.42	7.07	5.98	4.54	3.54	2.90	(44)	(44)
		RadStd	0.22	0.40	0.29	0.34	0.48	0.46	0.39	0.20	0.24	0.33	0.21	0.22	(45)	(45)

Historical Trends

		DBAvg	Heating		Cooling			Degree-Days				
			99% DB	99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3	
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	
Station Only		+0.79	N/A	N/A	+1.88	N/A	N/A	N/A	-158	+310	+170	(46)
Regional (50 neighbors)		+0.63	N/A	-1.70	+0.72	N/A	N/A	N/A	-112	+223	+112	(47)

Nomenclature: See separate page